

Antibiotic Susceptibility Testing

Laboratory Report — Kirby-Bauer Disk Diffusion Method

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Technique category: Clinical Microbiology

1. Objective

To determine the antibiotic susceptibility profile of three bacterial isolates against a panel of four antibiotics using the Kirby-Bauer disk diffusion method, and to classify each result as Susceptible, Intermediate, or Resistant.

2. Introduction

The disk diffusion method assesses how effectively an antibiotic inhibits bacterial growth. A standardised bacterial lawn is spread on Mueller-Hinton agar, and antibiotic-impregnated disks are placed on the surface. As the antibiotic diffuses outward, it creates a concentration gradient; after incubation, a clear zone of inhibited growth forms around effective disks. The diameter of this zone is inversely related to the minimum inhibitory concentration and is compared against standardised breakpoint tables specific to each antibiotic to classify susceptibility.

3. Materials

- Mueller-Hinton agar plates
- Bacterial isolates, standardised to 0.5 McFarland turbidity
- Antibiotic disks: Ampicillin, Ciprofloxacin, Gentamicin, Tetracycline
- Sterile cotton swabs and forceps
- Incubator (35–37°C), calipers for zone measurement

4. Method

- Standardised each bacterial suspension to 0.5 McFarland turbidity.
- Swabbed suspension evenly across the Mueller-Hinton agar surface (lawn inoculation).
- Placed antibiotic disks on the agar surface, spaced adequately apart.
- Incubated plates at 35–37°C for 16–18 hours.
- Measured the diameter of each zone of inhibition in mm.
- Compared measured zones against breakpoint tables to classify susceptibility.

5. Results

Twelve isolate-antibiotic combinations were tested across three isolates and four antibiotics. Three combinations were classified as Resistant, five as Intermediate, and four as Susceptible.

Isolate	Antibiotic	Zone (mm)	Classification
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Isolate_1	Ampicillin	8	Resistant
Isolate_1	Ciprofloxacin	24	Susceptible
Isolate_1	Gentamicin	16	Susceptible
Isolate_1	Tetracycline	20	Susceptible
Isolate_2	Ampicillin	19	Susceptible
Isolate_2	Ciprofloxacin	14	Resistant
Isolate_2	Gentamicin	13	Intermediate
Isolate_2	Tetracycline	12	Resistant
Isolate_3	Ampicillin	15	Intermediate
Isolate_3	Ciprofloxacin	18	Intermediate
Isolate_3	Gentamicin	15	Susceptible
Isolate_3	Tetracycline	17	Intermediate

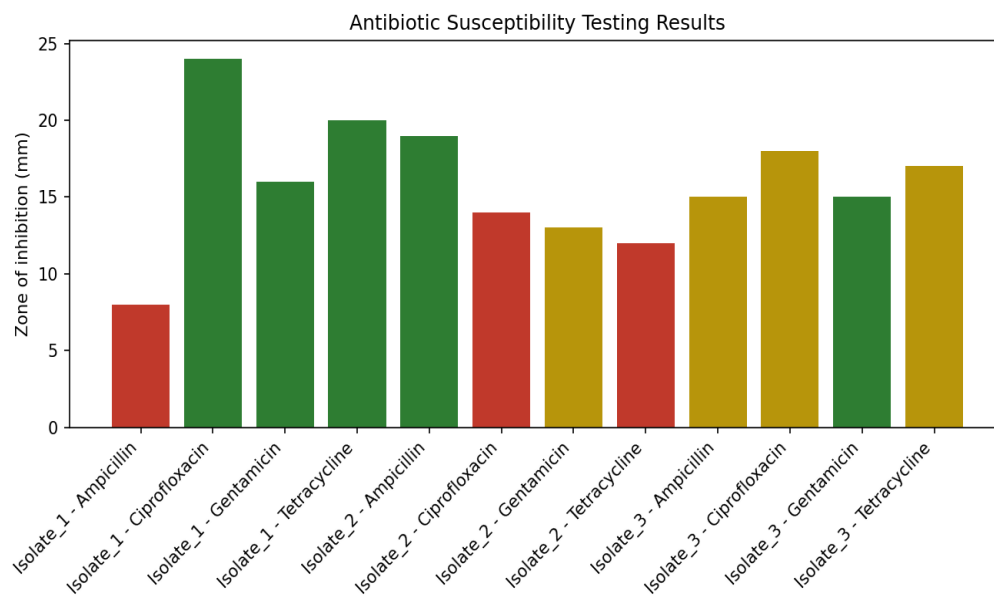


Figure 1. Zone of inhibition by isolate and antibiotic, coloured by susceptibility classification.

6. Discussion

Isolate_1 showed a resistant phenotype specifically to Ampicillin while remaining susceptible to the other three antibiotics tested, a pattern consistent with beta-lactamase production, a common resistance mechanism against penicillin-class antibiotics. Isolate_2 showed resistance or intermediate results across three of the four antibiotics tested, suggesting a broader resistance profile that would warrant further testing (e.g. extended antibiotic panel, resistance gene screening) before treatment selection. Isolate_3 showed no full resistance but several intermediate results, which in a clinical context would typically prompt either a higher dosing regimen or selection of an alternative agent with a clearer susceptible result, since intermediate classifications carry more uncertainty than a clear susceptible or resistant result.

7. Conclusion

This exercise demonstrated the complete Kirby-Bauer workflow, from zone measurement through breakpoint-based classification, and highlighted how susceptibility patterns can vary meaningfully between isolates and inform which antibiotics would be appropriate treatment choices for each.